1 Introduction and inequalities

Week 1 · Worksheet

Name _____ Date _____

Exercises

Show your work. Each exercise has a hint at the end: try without it first.

1. Classify by degree Warm-up · Lesson 1

Give the degree of each equation after simplifying: (a) $5x - 3 = 2x + 9$; (b) $x(x - 4) = 12$; (c) $(x - 1)(x^2 + 2) = 0$; (d) $x^2 + 3x = x^2 - 6$; (e) $2x^4 - x = 7$.

Hint: Expand the products and cancel equal terms on both sides.

2. A model rocket Warm-up · Lesson 1

A model rocket's height is $h = -2(t - 1)(t - 9)$ metres, t in seconds. It is fired from the ground at $t = 1$. When does it land?

Hint: On the ground means $h = 0$. Use the zero product rule.

3. Not what it looks like Practice · Lesson 1

Find the degree and solve $(x - 3)^2 = x^2 - 15$.

Hint: Expand the left side first.

4. Simplify the roots Warm-up · Lesson 2

Solve $2x^2 - 8x + 5 = 0$ and simplify the result.

Hint: $\Delta = 64 - 40 = 24$ and $\sqrt{24} = 2\sqrt{6}$.

5. The biggest garden Practice · Lesson 2

You have 40 m of fence for a rectangular garden against a wall (only three sides need fence). With width x, the area is $A(x) = x(40 - 2x)$. Which width gives the largest area, and how large is it?

Hint: $A(x) = -2x^2 + 40x$ opens down: find the vertex.

6. Best ticket price Challenge · Lesson 2

A school play sells 200 - 4p tickets when the price is p. Revenue is $R(p) = p(200 - 4p)$. Which price brings in the most money?

Hint: $R(p) = -4p^2 + 200p$. Vertex at $-b/2a$.

7. Braking distance Warm-up · Lesson 3

A rough rule estimates a car's speed from its skid marks: $v = \sqrt{(20d)}$, with v in m/s and d in metres. If $v = 20$ m/s, how long were the marks?

Hint: Square both sides.

8. Thermostat Practice · Lesson 3

A heater is set to 21 °C and reacts when the room is exactly 1.5 °C away from it: $|T - 21| = 1.5$. At which temperatures does it react?

Hint: $T - 21 = 1.5$ or $T - 21 = -1.5$.

9. Two printers Challenge · Lesson 3

One printer finishes a job in 6 hours. With a second printer helping, the job takes 4 hours: $1/6 + 1/x = 1/4$. How long would the second printer take alone?

Hint: $x \neq 0$. Multiply by $12x$.

10. Mobile data Warm-up · Lesson 4

A plan costs 12 a month plus 0.50 for each extra GB. This month's bill was 19.50. How many extra GB were used?

Hint: $12 + 0.5g = 19.5$.

11. A tablet screen Practice · Lesson 4

A 4:3 tablet screen has height $(3/4)\cdot w$ and an area of 192 cm². Find its width and height.

Hint: $w\cdot(3/4)w = 192$.

12. Saving for a trip Challenge · Lesson 4

Lucía has 120 saved and adds 35 each week. Tomás has 260 and adds 15 each week. After how many weeks will they have the same amount?

Hint: $120 + 35w = 260 + 15w$.

13. Three roots Practice · Lesson 5

Solve $x^3 - 7x - 6 = 0$ with Ruffini's rule.

Hint: Coefficients 1, 0, −7, −6; try $x = -1$.

14. A storage box Challenge · Lesson 5

A box has a square base of side x cm and height x − 2 cm. Its volume is 32 cm³: $x^2(x - 2) = 32$. Find x.

Hint: $x^3 - 2x^2 - 32 = 0$; try the divisors of 32.

15. A fourth-degree equation Challenge · Lesson 5

Solve $x^4 - 2x^3 - 7x^2 + 8x + 12 = 0$.

Hint: Try $x = -1$ first, then $x = 2$ on the quotient.

16. Rules as intervals Warm-up · Lesson 6

Write as intervals and in set notation: (a) a highway speed between 60 and 100 km/h, both allowed; (b) a child fare for ages under 12 (age ≥ 0); (c) a password of at least 8 characters.

Hint: Decide bracket or parenthesis at each end.

17. Brackets on both sides Practice · Lesson 6

Solve $4(x - 1) - 3 < 6x + 5$.

Hint: Expand, collect x, and watch the sign when you divide.

18. Gym or pay per class? Practice · Lesson 6

A gym charges 30 a month plus 2 per class; a studio charges 5 per class. From how many classes a month is the gym cheaper?

Hint: $30 + 2c < 5c$.

19. When is there profit? Practice · Lesson 7

A small bakery's daily profit is $P(x) = -x^2 + 14x - 40$, where x is hundreds of loaves. For which x is there profit ($P > 0$)?

Hint: Multiply by -1 and flip: $x^2 - 14x + 40 < 0 = (x - 4)(x - 10) < 0$.

20. Three factors Challenge · Lesson 7

A response-time model is $f(x) = (x^2 - 4)(x - 5)$. For which x is $f(x) \geq 0$?

Hint: $(x + 2)(x - 2)(x - 5)$: critical points -2, 2, 5.

21. Average cost Challenge · Lesson 7

Printing x flyers costs 200 plus 5 each, so the cost per flyer is $(200 + 5x)/x$. For which x is it at most 9?

Hint: Subtract 9: $(200 - 4x)/x \le 0$. Only $x > 0$ makes sense.

22. Fridge temperature Warm-up · Lesson 8

Vaccines must be kept at 5 °C with a margin of 3 °C: $|T - 5| \le 3$. Which temperatures are safe?

Hint: $-3 \le T - 5 \le 3$.

23. Two absolute values Practice · Lesson 8

Solve $|x - 2| + |x + 1| = 5$.

Hint: Cases: $x < -1$, $-1 \le x < 2$, $x \ge 2$.

24. An equation with a parameter Challenge · Lesson 8

For which values of m does $x^2 + mx + 9 = 0$ have two different real roots? One? None?

Hint: $\Delta = m^2 - 36$. Solve $\Delta > 0$, $\Delta = 0$, $\Delta < 0$.

25. A ticket budget Challenge · Lesson 8

Adult tickets cost 12 and child tickets 7; you can spend at most 84: $12x + 7y \le 84$ with $x, y \ge 0$. Can you buy 3 adult and 6 child tickets? And 4 and 6?

Hint: Substitute each point into the inequality.

Quick check

Circle one answer.

26. How many real roots has $4x^2 - 4x + 1 = 0$?

A) Two B) One (repeated) C) None

27. A drone's height is $h(t) = -4t^2 + 24t + 5$ metres. What is its maximum height?

A) 41 m, at t = 3 s B) 5 m, at t = 0 s C) 77 m, at t = 6 s

28. A classmate solved $x^2 = 16$ and wrote "x = 4". What is missing?

A) The root x = -4 B) Nothing C) The root x = 8

29. Which values solve $\sqrt{(x + 6)} = x$?

A) Only x = 3 B) x = 3 and x = -2 C) Only x = -2

30. A rectangle is 2 m longer than it is wide and has area 63 m^2. Solving $w(w + 2) = 63$ gives w = 7 or w = -9. What is the width?

A) 7 m B) -9 m C) Both

31. Which is a root of $x^3 - 2x^2 - 5x + 6 = 0$?

A) x = 1 B) x = -1 C) x = 2

32. Solve $6 - 2x \geq 10$.

A) $x \geq -2$ B) $x \leq -2$ C) $x \leq 2$

33. Printer A costs 120 plus 0.02 per page. Printer B costs 60 plus 0.05 per page. From how many pages is A cheaper?

A) More than 2,000 pages B) More than 600 pages C) Fewer than 2,000 pages

Answer key

1. (a) 1st ($x = 4$); (b) 2nd: $x^2 - 4x - 12 = 0$; (c) 3rd: $x^3 - x^2 + 2x - 2 = 0$; (d) 1st: $3x = -6$; (e) 4th.
2. $h = 0$ when $t = 1$ or $t = 9$. It lands at $t = 9$ s.
3. $x^2 - 6x + 9 = x^2 - 15 \rightarrow -6x = -24 \rightarrow x = 4$. It is first degree.
4. $x = (8 \pm 2\sqrt{6})/4 = (4 \pm \sqrt{6})/2$, about 3.22 and 0.78.
5. $x = -40/(2\cdot(-2)) = 10$ m, and $A(10) = 10\cdot 20 = 200$ m².
6. $p = -200/(-8) = 25$. They sell 100 tickets and $R = 2{,}500$.
7. $400 = 20d \rightarrow d = 20$ m. Check: $\sqrt{400} = 20$.
8. $T = 22.5$ °C or $T = 19.5$ °C.
9. $2x + 12 = 3x \rightarrow x = 12$ hours.
10. $0.5g = 7.5 \rightarrow g = 15$ GB.
11. $w^2 = 256 \rightarrow w = 16$ cm, height 12 cm.
12. $20w = 140 \rightarrow w = 7$ weeks; both will have 365.
13. Quotient $x^2 - x - 6 = (x - 3)(x + 2)$. Roots $-1, 3, -2$.
14. $x = 4$ ($64 - 32 - 32 = 0$). The quotient $x^2 + 2x + 8$ has $\Delta < 0$, so 4 cm is the only answer: a $4 \times 4 \times 2$ box.
15. $x = -1$ leaves $x^3 - 3x^2 - 4x + 12$; $x = 2$ leaves $x^2 - x - 6 = (x - 3)(x + 2)$. Roots $-1, 2, 3, -2$.
16. (a) $[60, 100] = \{v : 60 \le v \le 100\}$; (b) $[0, 12) = \{a : 0 \le a < 12\}$; (c) $[8, \infty) = \{n : n \ge 8\}$.
17. $4x - 7 < 6x + 5 \rightarrow -2x < 12 \rightarrow x > -6$: $(-6, \infty)$.
18. $30 < 3c \rightarrow c > 10$: from 11 classes on.
19. $(4, 10)$: between 400 and 1,000 loaves. At 4 and 10 the profit is exactly 0.
20. $[-2, 2] \cup [5, \infty)$. Tests: -3 gives −, 0 gives +, 3 gives −, 6 gives +.
21. The table gives $(-\infty, 0) \cup [50, \infty)$; with $x > 0$, from 50 flyers on.
22. $[2, 8]$ °C.
23. $x < -1$: $-2x + 1 = 5 \rightarrow x = -2$ ✓. Middle: $3 = 5$, impossible. $x \ge 2$: $2x - 1 = 5 \rightarrow x = 3$ ✓. Solutions $\{-2, 3\}$.
24. Two roots: $m < -6$ or $m > 6$. One: $m = \pm 6$. None: $-6 < m < 6$.
25. (3, 6): $36 + 42 = 78 \le 84$, yes. (4, 6): $48 + 42 = 90 > 84$, no: that point is outside the shaded region.
26. **B** — $\Delta = 16 - 16 = 0$. In fact $4x^2 - 4x + 1 = (2x - 1)^2$, so $x = 1/2$: the parabola touches the x-axis once.
27. **A** — $t = -24/(2\cdot(-4)) = 3$; $h(3) = -36 + 72 + 5 = 41$.
28. **A** — $(-4)^2 = 16$ as well: $x = \pm 4$.
29. **A** — Squaring gives $x^2 - x - 6 = 0$, so 3 or -2; but $\sqrt{4} = 2 \ne -2$, so -2 is extraneous.
30. **A** — A width cannot be negative, so discard -9.
31. **A** — $1 - 2 - 5 + 6 = 0$. (For -1: $-1 - 2 + 5 + 6 = 8$; for 2: $8 - 8 - 10 + 6 = -4$.)
32. **B** — $-2x \ge 4$; dividing by -2 flips the sign: $x \le -2$.
33. **A** — $120 + 0.02p < 60 + 0.05p$ gives $60 < 0.03p$, so $p > 2000$.